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Creators of BloodHound

Attack Paths in Practice

Okta, GitHub, and Jamf with BloodHound Enterprise

Justin Kohler
Chief Product Officer

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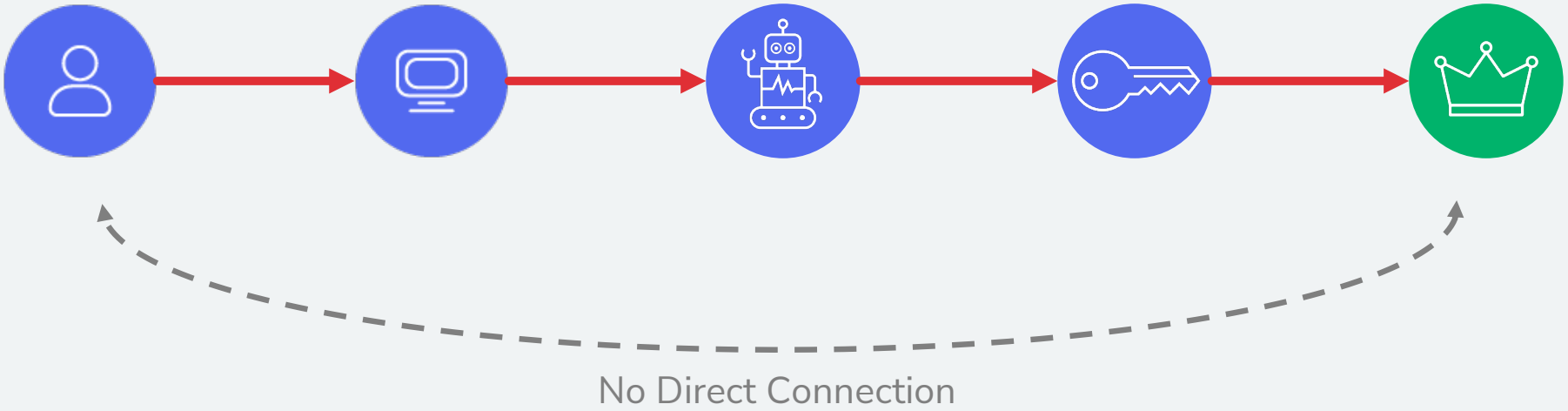
Chief Technology Officer

Attack paths are how privileges and behavior connect identities to control



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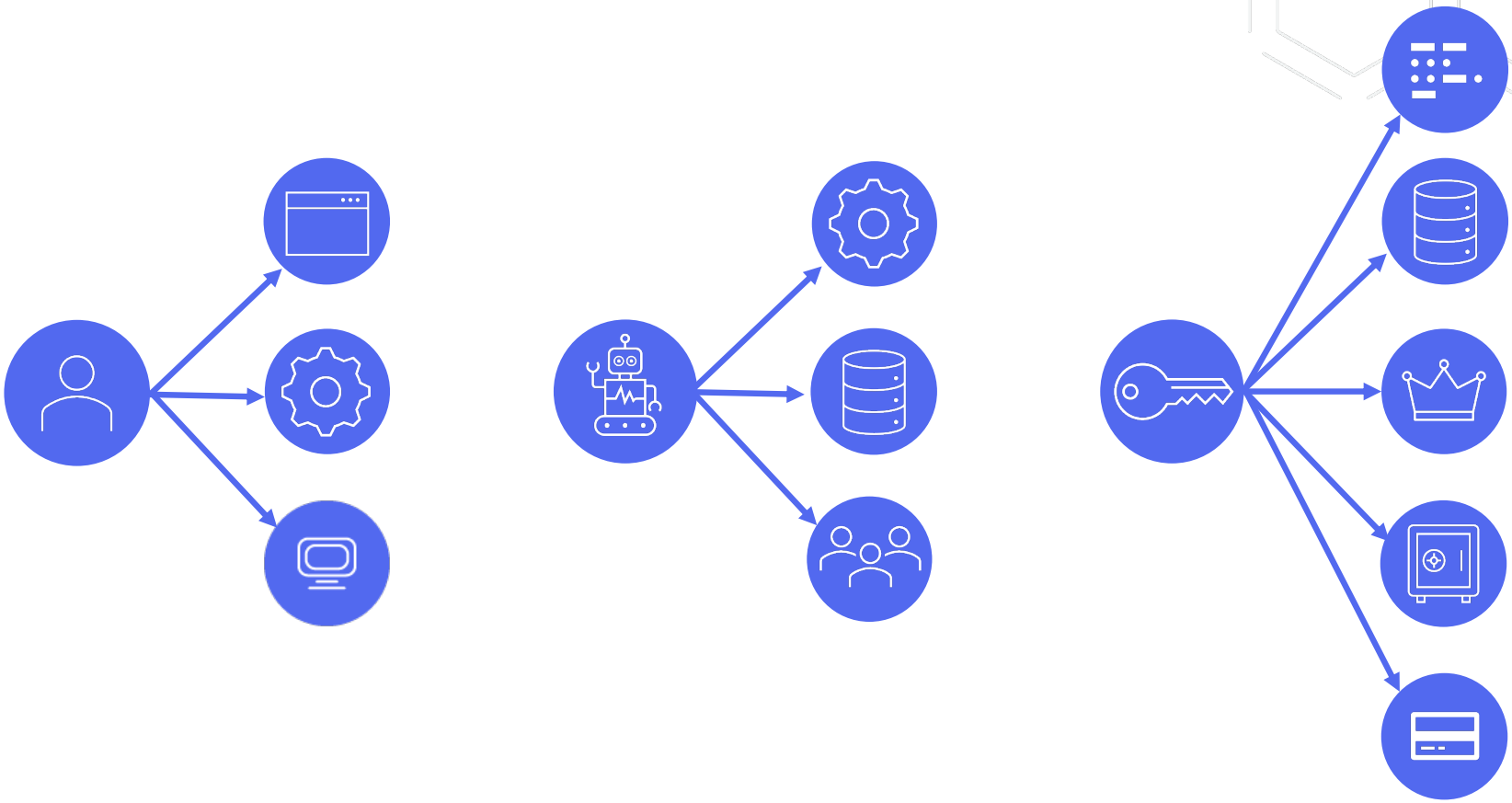
Benign, isolated identity configurations



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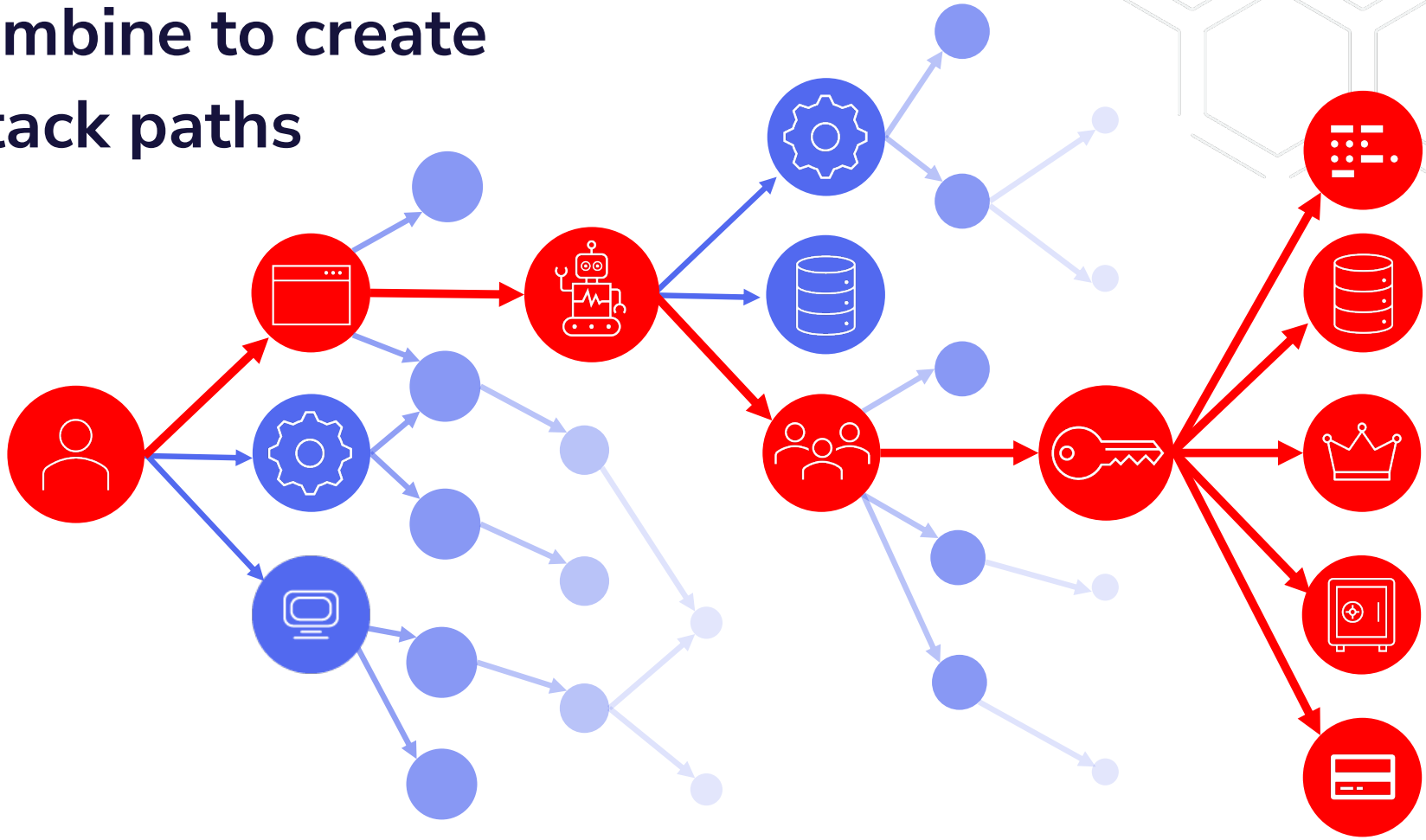
Combine to create attack paths



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From Bad to Ugly:
Cross-Platform Attack Paths

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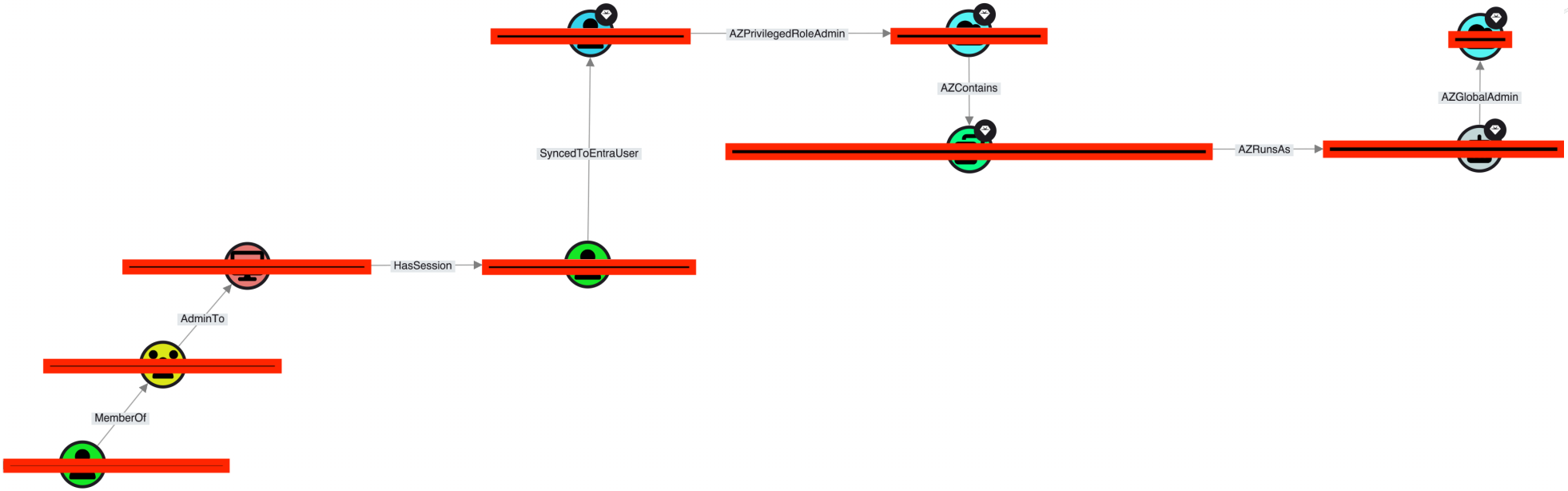
2 Years Ago: BloodHound visualizes hybrid MSFT Attack Paths



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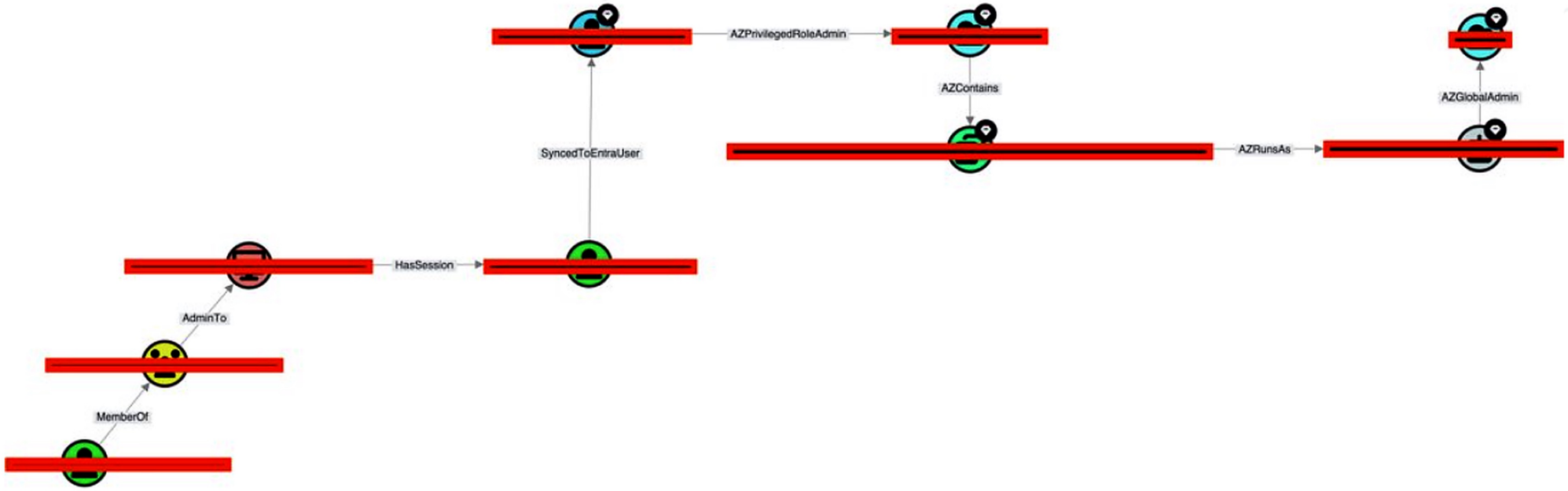
2 Years Ago: BloodHound visualizes hybrid MSFT Attack Paths



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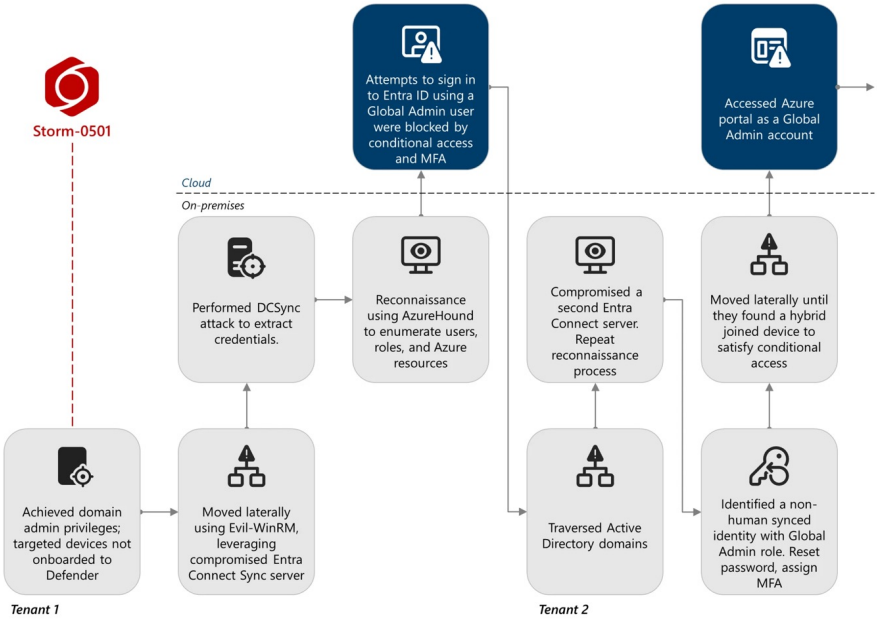
Storm-0501's evolving techniques lead to cloud-based ransomware



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From CI/CD to cloud compromise: Real-world breach using OpenID Connect abuse

Day 1
Initial Compromise

- Unknown threat group compromised the popular 'Nx' package
- Victim organization's developer updated existing NPM packages, including compromised 'Nx' package that executed malicious code
- QUIETVAULT malware obtained GitHub and NPM API keys from developer's workstation using AI that assisted with identifying files of interest
- QUIETVAULT malware uploaded API keys to public GitHub repository

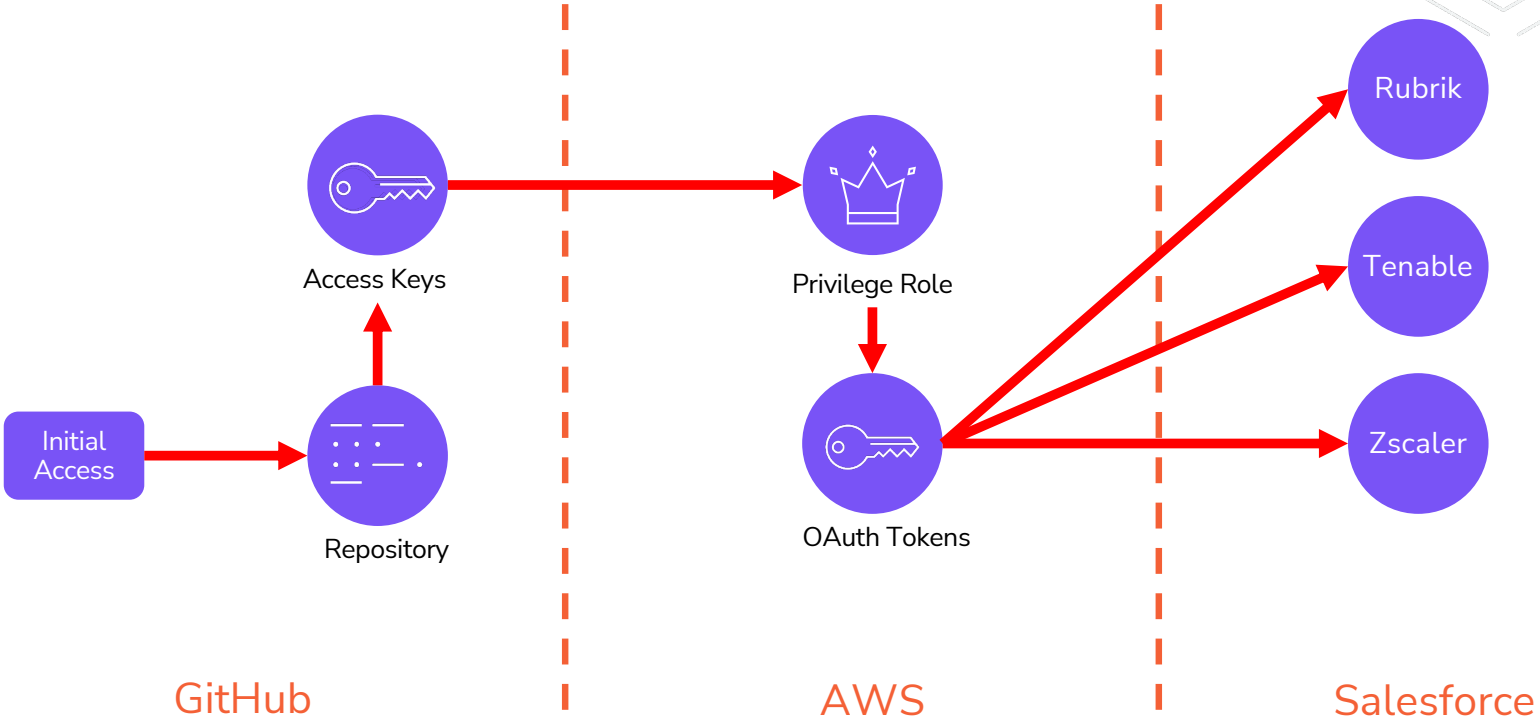
Day 3
Escalate Privilege

- UNC6426 accessed victim organization's GitHub repository using stolen keys
- UNC6426 enumerated victim organization's GitHub environment
- UNC6426 abused CI/CD pipeline, gaining access to AWS API keys

Day 4
Complete Objectives

- UNC6426 accessed victim organization's AWS environment and created unauthorized administrative credentials to escalate privileges
- UNC6426 exfiltrated data from victim organization's S3 buckets, renamed internal GitHub repositories and made them publicly accessible

SalesLoft Drift Breach Attack Path



Source: Widespread Data Theft Targets Salesforce Instances via Salesloft Drift

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Trivy Security Scanner GitHub Actions Breached, 75 Tags Hijacked to Steal CI/CD Secrets

"We identified that an attacker force-pushed 75 out of 76 version tags in the aquasecurity/trivy-action repository, the official GitHub Action for running Trivy vulnerability scans in CI/CD pipelines," Socket security researcher Philipp Burckhardt said. "These tags were modified to serve a malicious payload, **effectively turning trusted version references into a distribution mechanism for an infostealer**."

The payload executes within GitHub Actions runners and aims to **extract valuable developer secrets from CI/CD environments, such as SSH keys, credentials for cloud service providers, databases, Git, Docker configurations, Kubernetes tokens, and cryptocurrency wallets.**

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GitHub, Okta, and Jamf

Enterprise Extensions now in BloodHound Enterprise

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Enterprise Extensions

- Automated Choke Point Analysis
- Prioritization and trend analysis
- Built-in Remediations
- Continuous monitoring & reconciliation
- Authored and supported by SpecterOps

Open-Source XHounds

- Authored and supported by the community
- Open-Source: Use at your own risk

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Why we chose these first three

Top IDP for most organizations.

- *If we can control the IDP, we can get access anywhere.*

Used to manage macOS fleets

- *If we can control the management plane, we can control everything*

Holds source code, company IP

- *With access to GitHub, can we manipulate source code or steal IP*

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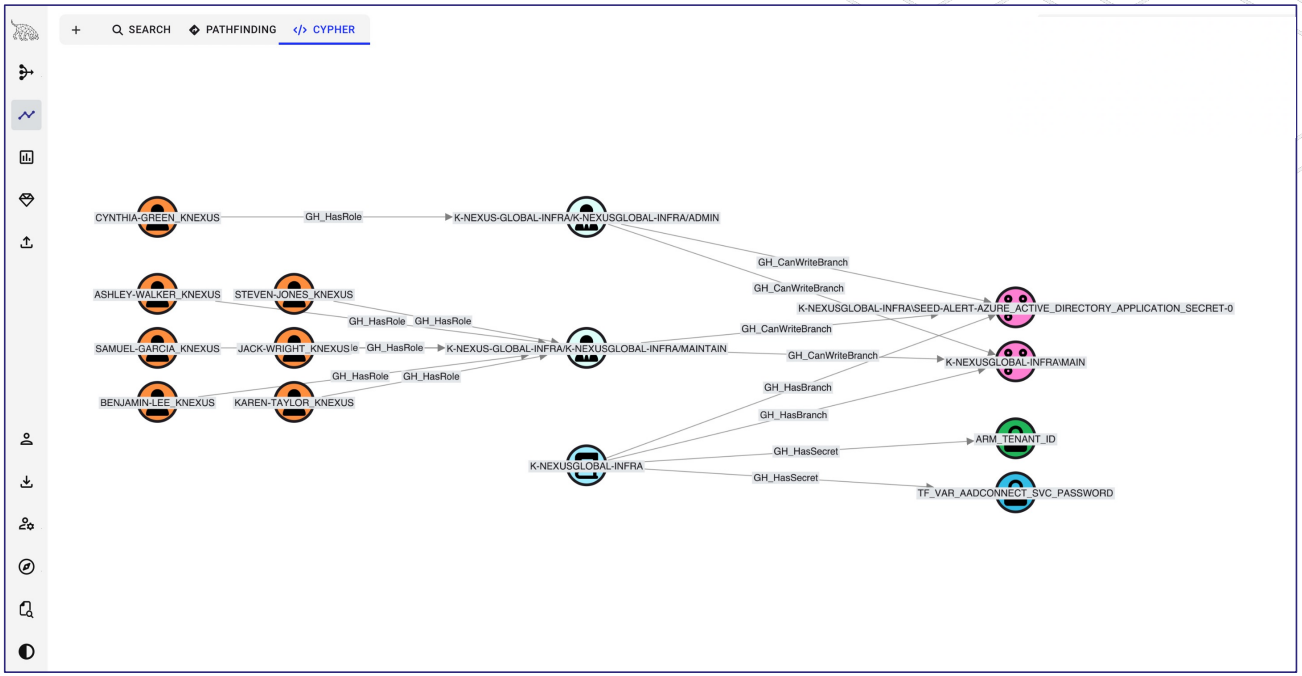
Attack Paths in GitHub with BloodHound Enterprise

GitHub

- Identity access to secrets
- Unprotected branches
- Risky branch protections
- Branch protection bypasses
- Self-approval workflows
- Public repos
- Privileged custom roles
- PAT access to all repos



Branch Protection Bypasses -> Secret Exfiltration

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Targeting Secret Scan Alerts in GitHub

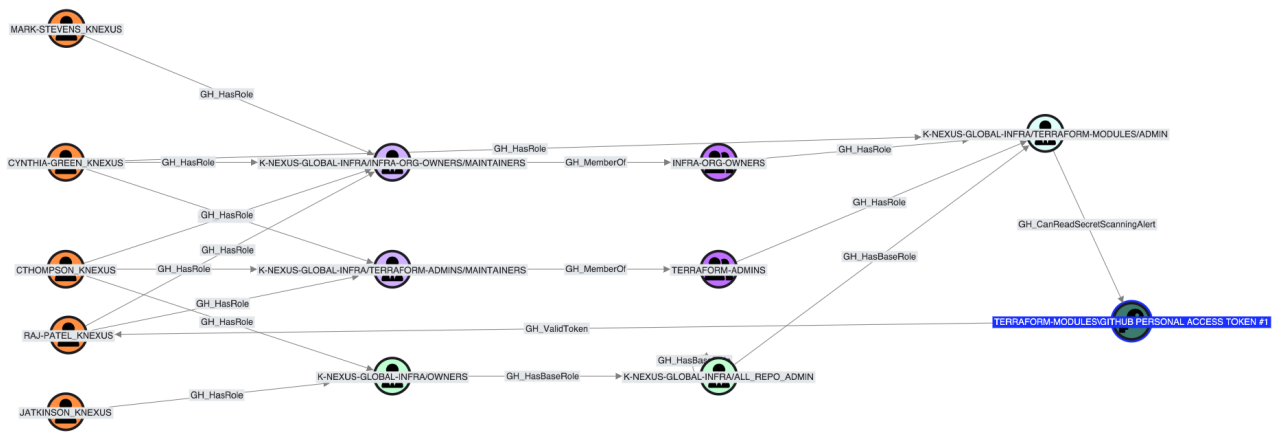
SEARCH | PATHFINDING | CYPHER

Saved Queries

Auto-run selected query

```
1 MATCH p=(role:GH_Role)-
  [:GH_HasRole|GH_HasBaseRole|GH_MemberOf|GH_ReadRepoCo
  ntents*1..]->(reporole:GH_Role)-
  [:GH_CanReadSecretScanningAlert]->
  (n:GH_SecretScanningAlert
  {objectid:'5353415F4F5F6B67444F445F575569415F525F6B67
  444F526C67324A675F31'})-[]->()
2 MATCH p1=(role)<-[:GH_HasRole]-(:GH_User)
3 OPTIONAL MATCH p2=(reporole)<-[:GH_HasRole]-
  (:GH_User)
4 RETURN p,p1,p2
```

Tag | Save | Run



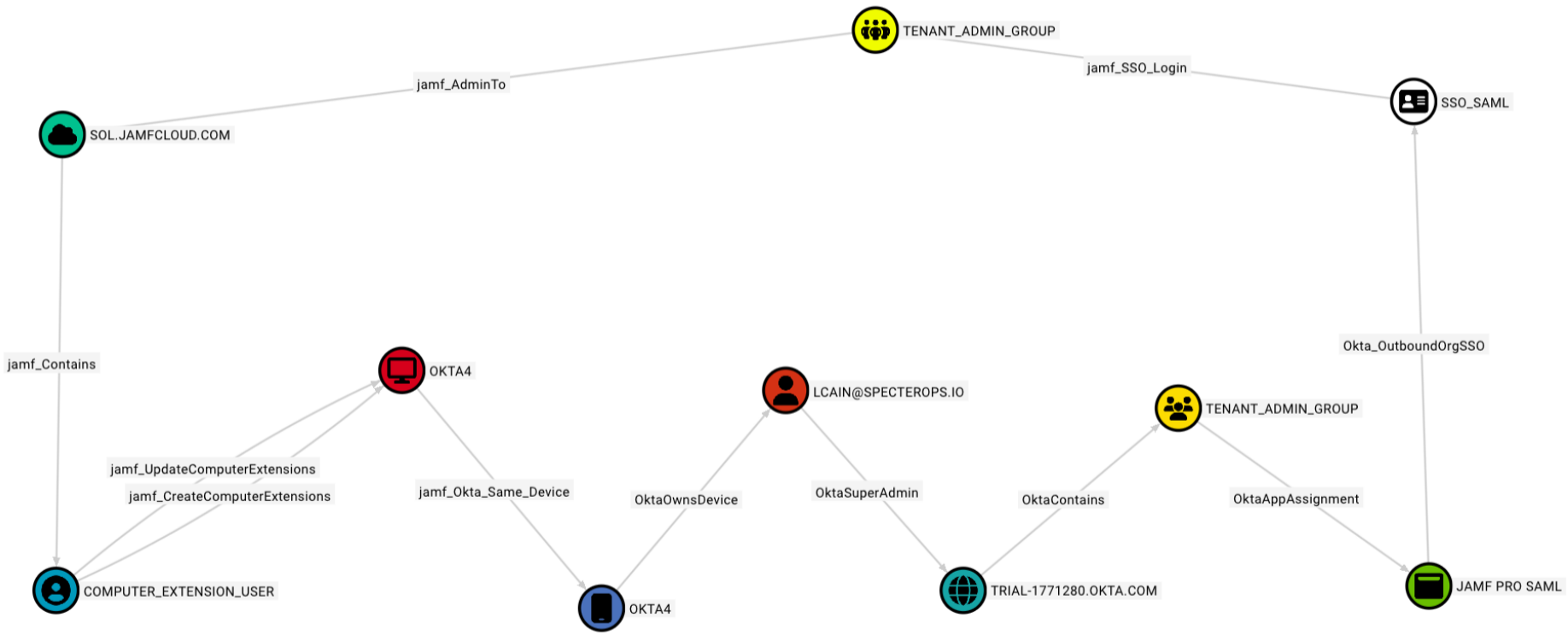
Hide Labels | Layout | Export | Search

TERRAFORM-MODULES\GITHUB PERSON...

Object Information

Node Type:	GH_SecretScanningAlert
Object ID:	5353415F4F5F6B67444F445F575569415F525F6B67444F526...
Created At:	2026-03-13T16:57:45Z
Environment Name:	k-nexus-global-infra
Environmentid:	4F5F6B67444F445F57556941
Last Seen by BloodHound:	2026-03-19 22:14 PDT (GMT-0700)
Node Id:	5353415F4F5F6B67444F445F575569415F525F6B67444F526...
Primary Kind:	GH_SecretScanningAlert
Query Alert Viewers:	MATCH p=(role:GH_Role)-[:GH_HasRole|GH_HasBaseRole|GH_MemberOf|GH_ViewSecre...>(:GH_Repository)-[:GH_Contains]->(:GH_SecretScanningAlert {node_id:'SSA_4F5F6B67444F445F57556941_525F6B67444F5... MATCH p1=(role)<-[:GH_HasRole]-(:GH_User) RETURN p,p1
Query Repository:	MATCH p=(r:GH_SecretScanningAlert {node_id:'SSA_4F5F6B67444F445F57556941_525F6B67444F5... <-[:GH_Contains]-(repo:GH_Repository) RETURN p
Repository Id:	R_kgDORlg2Jg
Repository Name:	terraform-modules
Repository Url:	https://github.com/k-nexus-global-infra/terraform-modules
Secret Type Display Name:	GitHub Personal Access Token
Secret Type:	github_personal_access_token
Secret:	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
State:	open
Updated At:	2026-03-13T19:21:05Z
Url:	https://github.com/k-nexus-global-infra/terraform-modules/security/secret-scanning/1
Validity:	active

macOS fleet compromise through okta & jamf



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Trivy breach from GitHub to AWS

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Attack Paths across platforms uncovers new problems

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An Enterprise-wide call to action

The rise of the "Messy Middle"

- AD and Entra ID may be separate teams but likely sister organizations
- Who is responsible for the Attack Path that crosses from GitHub?

Laddering Privilege across Platforms

- Escalating Privilege through one platform may be impossible
- But pivoting back and forth to incrementally gain privilege is real

Clean-Source Principal

- "All security dependencies must be as trustworthy as the object being secured"

2:15 PM Monday
Hybrid Attacks and Identities in Motion: Anatomy of Modern Attack Paths
Elad Shamir VP of Research, SpecterOps
Will Schroeder Security Researcher, SpecterOps

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Key takeaways

1. You can’t secure what you can’t see
2. Every connection introduces risk
3. Attack Paths are inevitable, unmanaged ones are optional

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Thank you

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State of Identity Attack Path Management

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